

AI-based Urine Microscopy Image Analysis for Predicting Urinary Tract and Renal Diseases

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Abstract— The manual microscopic examination of urine is a crucial step in diagnosis of urinary tract and renal diseases. However, the process is time-consuming and operator dependent. Most of the existing automated urinalysis techniques only consider the microscopic examination of individual components or the black-box-based prediction models. There is a lack of a comprehensive framework that incorporates the microscopic examination of the microscopic components with the clinical diagnostic logic. In this regard, the present work proposes an artificial intelligence-based urinalysis system for the microscopic examination of the components in the urine sample to generate diagnostic outcomes. In the proposed system, the microscopic components like white blood cells, red blood cells, bacteria, yeast, crystals, and casts are detected and analyzed to generate the diagnostic outcomes for the diagnosis of urinary tract infection, kidney stone risk, hematuria causes, and casts-related renal diseases. In the proposed system, efficient lightweight models ensure precision and effectiveness in identifying various biological entities. White blood cells are detected with a mAP@0.5 score of 0.96, yeast with over 0.94, and crystals with more than 0.91, yielding a classification accuracy of 99.41% for crystals. The system detects microscopic elements like casts with a mAP@0.5 score of 0.80. The system also incorporates auxiliary clinical data to enhance diagnostic results for various diseases.

Keywords— automated urinalysis, clinical decision support, urine microscopy, urinalysis system, urinary tract infection

I. INTRODUCTION

In clinical medicine, urinalysis is a fundamental diagnostic technique that yields important information regarding kidney diseases and urinary tract infections (UTIs). Important particles such as white blood cells (WBCs), red blood cells

(RBCs), bacteria, yeast, casts, and crystals can be found by looking at the urine sediment under a microscope [1]. The presence, shape, and number of these particles can indicate infection, inflammation, blood in urine, kidney stones, and other renal disorders [2], [3], [4]. Despite its relevance, traditional manual urine microscopy requires a lot of time and effort, and it strongly depends on the operator, which leads to variability in results and consistency [5].

The examination of microscopic images could be completely automated thanks to recent advancements in artificial intelligence (AI) technology, especially in deep learning-based image analysis. Particles observed in urine sediment are among the complex visual patterns in medical images that Convolutional Neural Networks (CNNs) and object identification models like You Only Look Once (YOLO) have shown good accuracy in identifying, segmenting, and classifying [6], [7], [8], [9]. Nevertheless, the majority of automated urinalysis systems in use today only identify individual particles or forecast a particular disease result, losing the opportunity to develop a practical and all-encompassing diagnostic tool [4], [10]. Furthermore, there hasn't been much research done on integrating particle-level microscopic information with disease-specific clinical reasoning [5].

The goal of this project is to develop a comprehensive AI-based framework for evaluating microscope images in order to forecast and describe kidney and urinary tract disorders. The suggested system will be a modular but coherent pipeline that analyzes various urine sediment components using a number of deep learning models, each of which aims to achieve a particular diagnostic objective. This project aims to reduce human burden, increase diagnosis accuracy, and

facilitate faster, data-driven decisions in clinical settings by combining targeted interpretation for diseases with comprehensive particle detection, segmentation, and classification [5], [6], [8].

There are four interrelated components to the project. The first section focuses on applying AI to diagnose UTIs by identifying WBCs, yeast, and bacteria in urine microscopy pictures. In order to improve interpretation and dependability in the diagnosis of infections, this attempts to integrate particle-level evidence with clinical data [6], [10], [11], [12]. Using a two-stage deep learning architecture for crystal localization and thorough subtype classification, the second section focuses on automating the detection and categorization of urine crystals to provide individualized estimates of kidney stone risk [4]. The third section focuses on urine cast detection and classification using AI, which aids in the differentiation of clinically significant cast subtypes that are crucial for the diagnosis of renal disorders [2], [3]. The fourth section focuses on using AI to identify and categorize RBCs, enabling the differentiation. In order to distinguish between glomerular and non-glomerular sources of blood in urine, the fourth section concentrates on identifying and categorizing RBCs using AI [1], [13].

By combining these several studies into a single AI-driven framework, this study seeks to further automated urine microscopy analysis beyond particle detection. The strategy positions AI-assisted urinalysis as a dependable support system for early identification and monitoring of urinary tract and kidney diseases by emphasizing clinical relevance, interpretability, and scalability [5], [11], [14].

II. LITERATURE REVIEW

Microscopy of urine remains a foundation in the diagnosis of urinary tract and renal diseases, as it allows for the direct observation of diagnostically important particles including white blood cells, red blood cells, bacteria, yeast, casts, and crystals, reflecting infection, inflammation, hematuria, and renal pathology. Recent advances concerning deep learning have significantly improved automated urine sediment analysis. CNN-based object detectors like YOLO and Bidirectional Context Propagation Networks showed pretty good performance in detecting WBCs and other formed elements in microscopic images [6], [7], [9], [15]. Image quality enhancement techniques have also been explored in medical imaging applications such as kidney stone detection in CT images highlighting the importance of image preprocessing in diagnostic pipelines [16]. Increasingly, large and clinically annotated datasets have accelerated these developments further [5]. In most of these studies focusing on detection, there has been a major focus on high accuracy of localization and computational efficiency, while considering limited clinical interpretation or diagnostic reasoning.

In the context of instance and pixel-level segmentation, various approaches including YOLOv5n-based models have been developed to further refine the boundary segmentation of particles and quantitative analysis [8]. Although the precision of the segmentation has been improved by these models, the result of the segmentation itself has not been

considered in higher-level diagnostic decision mechanisms based on the translation of microscopic-level results into disease-level diagnostics [4], [10]. Similarly, while various deep learning-based models have reported highly accurate bacterial classifications, they have been primarily reported from culture images and isolated bacterial samples rather than the microscopic examination of urine samples, which indicates a limitation in the real-world applicability of the approach in routine urinalysis examinations [17], [18].

Apart from microscopy, various articles aimed to predict UTIs with symptom-oriented UTI prediction models based on clinical features such as laboratory results, flow cytometry, optical techniques, and patient symptoms, which relied on machine learning algorithms [11], [14], [15], [17], [18], [19], [20]. Even though such systems possess strong predictive power, they remain black boxes without microscopic interpretability, hence hindering a clinician's trust. In addition, computerized analysis of urinary crystals, RBCs, and casts with deep learning has shown early indications of its potential to be used in kidney stones risk prediction, hematuria etiology prediction, and kidney disease diagnosis [2], [3], [4], [20], [21], [22], [23]. However, many such systems analyze each urine component separately without considering the sub-classifications within each component.

Overall, while the existing literature provides approaches to the detection, segmentation, classification, and prediction of urine sediment images, it mainly focuses on treating these as separate tasks. Nonetheless, as already mentioned, there is a gap, particularly with regards to the development of integrated, microscopy-based AI methodologies for the comprehensive diagnosis of disorders of the urinary tract and kidneys.

III. METHODOLOGY

A. AI-based Identification Of WBCs, Yeast and Bacteria In Urine Microscopy Images For UTI Classification

This project centers on the deep learning part based on microscopy of the proposed automated UTI diagnosis system. A task-specific deep learning model is applied in a modular, specialized-model approach to examine each type of urine sediment particle. Preprocessing steps such as scaling, normalization, and data augmentation were used to create distinctive datasets for bacteria, yeast cells, and white blood cells (WBCs).

A YOLOv11-n model, selected for its superior small-object detection ability and real-time processing capability, was used to perform WBC detection, which was implemented as a single-class object detection task. To provide accurate morphological details, WBC segmentation was carried out using a UNet++ model with an EfficientNet-B0 encoder that was trained on pixel-level WBC masks.

Yeast analysis was done in a similar manner. Yeast cells were detected using a YOLOv11-based detector trained on yeast-cell-specific bounding box annotations, and yeast segmentation was implemented using SegFormer-B2(MiT-B2) to enhance robustness in dense regions and refine boundaries.

Because there were no accurate bounding box annotations for bacteria, bacterial analysis was implemented as a classification problem. An EfficientNet-B0 classifier was trained to correctly identify *Escherichia coli* from non-bacterial samples using cropped urine microscopy images. All model outputs were collected and inputted into a rule-based decision layer for the following classification of UTIs.

Bacterial detection and segmentation are being developed at present using the modular structure of the proposed system and will be added to the project in the future once accurate bounding box annotations are made available.

The WBC cast detection and red blood cell (RBC) counts were obtained from supplementary modules developed by other team members and integrated into the proposed methodology. The counts of detected particles were merged using medically informed rules to forecast the types of UTIs, such as lower bacterial UTI (WBC + bacteria), upper bacterial UTI (WBC + bacteria + WBC casts), lower fungal UTI (WBC + yeast), upper fungal UTI (WBC + yeast + RBC), and no UTI with minimal or no cellular results.

Simultaneously, a machine learning-based symptom UTI prediction module was developed. Here two separate machine learning models were developed based on metadata. A symptom model offered a weak probabilistic prior, and a structured clinical indicator model facilitated UTI localization. Instead of combining datasets, a decision-level fusion approach was used to maintain the integrity of the datasets.

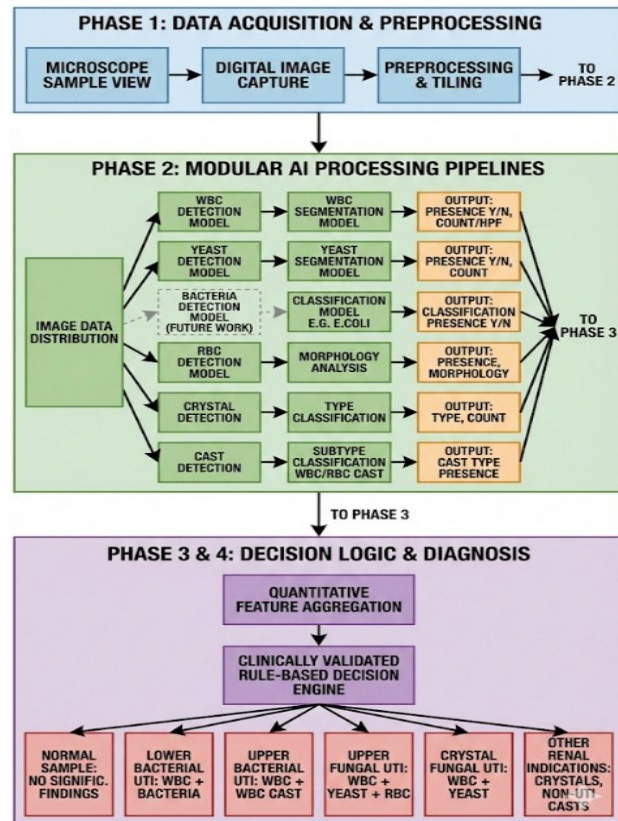


Fig. 1. End-to-end AI-based framework for automated urine microscopy diagnosis.

B. AI-Based Detection and Classification of Urinary Crystals for Personalized Prediction of Kidney Stone Risk

A two-stage deep learning technique has been applied to examine the urine crystals with the aim of identifying and classifying the major types of crystals associated with the formation of kidney stones. This study has concentrated on uric acid, phosphate, calcium oxalate dihydrate, and calcium oxalate monohydrate crystals. Datasets were used from publicly available images of microscopic views, split into training, validation, and testing sets, with a ratio of 80:10:10.

In the localization and detection of the crystal, the YOLO-based object detection model has been used. In the detection of small objects within the microscopic images, the YOLO11m architecture has been used, as it strikes a proper balance between detection accuracy and efficiency. All images were resized to 640 x 640 pixels. In the training of the network, the weights are initialized with the pretrained weights. Then, the weights are refined with the AdamW optimizer with a weight decay of 0.01. The initial learning rate was set at 5×10^{-4} , and the use of a cosine learning rate scheduler was made to ensure smooth convergence. The training was done for 50 epochs with a batch size set at 16. For avoiding overfitting, early stopping with patience set at 30 epochs was implemented, and the best set of weights was used for final evaluation.

In the second stage, the crystal regions that were identified were clipped and subjected to in-depth crystal classification through a convolutional neural network. In this case, all the photos were resized to 224 by 224 pixels to fit the requirements needed by the classification network. In addition to that, data augmentation techniques were applied to make the network more robust to prevent overfitting.

A convolutional neural network based on EfficientNet-B0 is used in the classification of crystals since it strikes a balance between computational efficiency and accuracy. Pre-trained weights are used to set up the network using the ImageNet dataset. In addition to that, the network is set to be frozen during the training process. Global average pooling, batch normalization, and dropout are applied in the feature extraction process. Lastly, the fully connected dense layer with the softmax activation function is applied in the multi-class classification process. The Adam optimizer is applied in the training of the classification model with the sparse categorical cross-entropy loss function and a learning rate of 1×10^{-3} . The training process is carried out through 30 epochs with a batch size of 32. Using a patience of three, early stopping is applied to the training process with the use of the validation loss. Checkpointing is also applied in the training process to retain the optimal weights in the model.

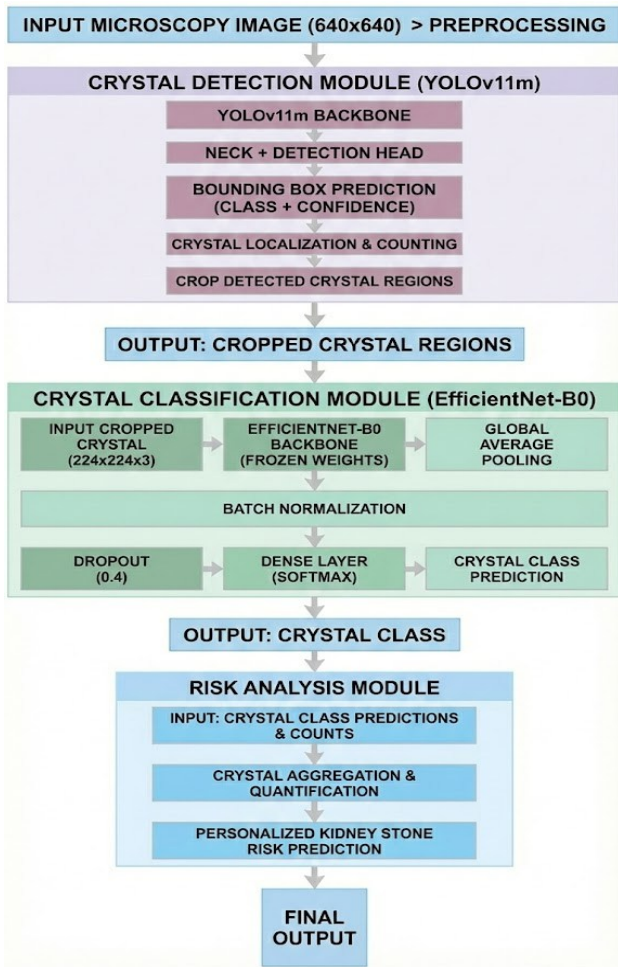


Fig. 2. Architecture of the proposed AI-based system for urinary crystal detection, classification, and kidney stone risk assessment

C. AI-based Detection Of RBCs In Urine Microscopy Images For Hematuria Cause Prediction

A two-tier system was developed for the automated analysis of red blood cell (RBC) morphology based on microscopic images of urine. In the first tier, the YOLOv8 model was used for the detection of individual RBCs from the microscopic images. According to the detected boxes, single RBC images were cropped for further detailed analysis with minimum background noise. All the cropped images were checked for the presence of single, intact RBCs, and the images were discarded for further analysis if they were found to be of poor quality.

After validation, the individual RBC images were retained for further processing and resizing as per the input requirement for the classification model. Normalization was done with ImageNet standards, and light data augmentation was carried out for the training process with random rotations and flipping for the model. Approximately 2,000 single RBC images were used for training, validation, and testing sets.

An EfficientNet-based CNN has been employed for the classification task. In the current study, the model EfficientNet-B3 has been selected because of its balanced complexity and performance. In addition, the model was

applied with pre-trained weights on the ImageNet dataset and fine-tuned for 25 epochs with a focus on recall for the binary classification of isomorphic and dysmorphic RBCs, which better correspond to the screening requirements in the clinical setting.

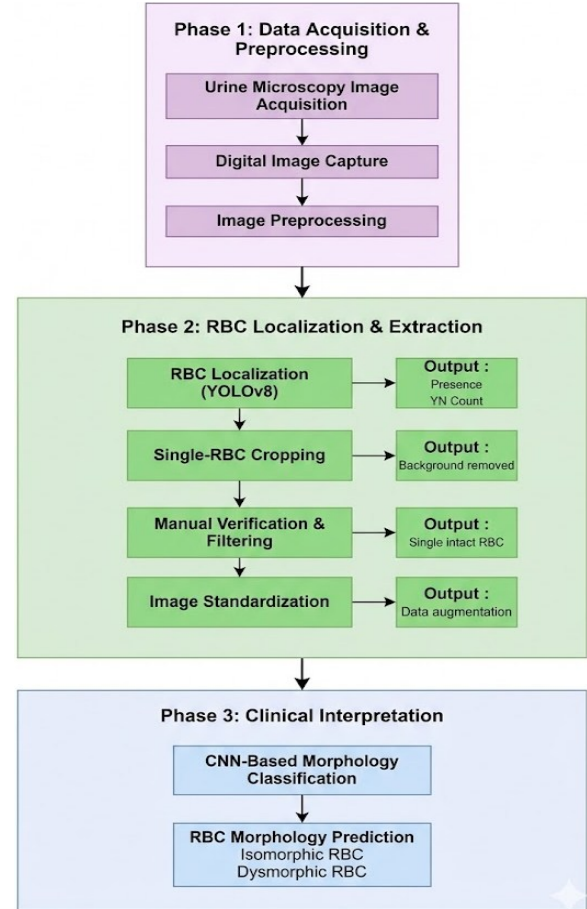


Fig. 3. Model architecture for automated RBC morphology classification from urine microscopy images.

D. AI-Based Detection and Classification of Urinary Casts for Kidney Disease Diagnosis

A dataset that was curated with a structured pipeline made with the Roboflow platform gave images of the urine sediment, which contained 5,391 samples with various objects such as urinary casts, along with other cells such as red blood cells, white blood cells, crystals, and other debris. The identification and categorization of the objects in the urine cast images were done with a two-stage method due to the intricacy of the objects in the images and the various visual patterns that could be observed in the microscope images. The accuracy, precision, and complexity of object detection models such as YOLOv8n, YOLOv8m, YOLOv10b, and RT-DETR were evaluated. Among the object detection models, the YOLOv8m model showed better accuracy and precision with moderate complexity. The YOLOv8m object detection model was selected for the identification of objects in the microscope images, which contained the objects such as urinary casts. In order to identify the objects such as urinary casts in the microscope images, the YOLOv8m object

detection model was trained with manually annotated bounding boxes. The dataset was split into test, validation, and training sets at an 80:10:10 ratio.

1,436 pictures with just the cast items were obtained by utilizing the trained model. Five cast subtypes of medical significance were manually recognized from the cropped images, namely granular cast (338 images), hyaline cast (320 images), red blood cell cast (66 images), white blood cell cast (182 images), and waxy cast (22 images). To mitigate class imbalance in rare urinary cast subtypes such as waxy casts, targeted data augmentation techniques including rotation, brightness variation, flipping, and Gaussian noise injection were applied to increase sample diversity and improve model robustness. Furthermore, as a potential future enhancement, GAN-based synthetic data generation could be explored to further improve performance on underrepresented classes. Four CNN architectures, namely ResNet-18, MobileNet-V2, EfficientNet-B0, and EfficientNet-B2, were used by the proposed system for the subtype classification problem. These architectures were initialized with pre-trained weights on the ImageNet dataset and fine-tuned on the set of cropped images of the casts, resized to a fixed spatial size of 224 x 224 pixels.

In order to handle the class imbalance problem of the five cast subtypes of medical significance, the proposed system used stratified five-fold cross-validation, inverse class weighting, and the macro-average evaluation metric for the subtype classification problem. The EfficientNet-B2 architecture, which was trained on the set of images of the casts, utilizing the Adam optimizer, weight decay, learning rate scheduling, and early stopping based on the validation set's macro-F1 score metric, showed the most consistent performance among the tested classifiers of the proposed system and was used as the subtype classifier of the proposed system. In order to recognize, count, and classify the subtypes of the urinary casts from the images of the microscopic examination of medical importance, the YOLOv8m detector and EfficientNet-B2 classifier were integrated into the proposed system utilizing FastAPI.

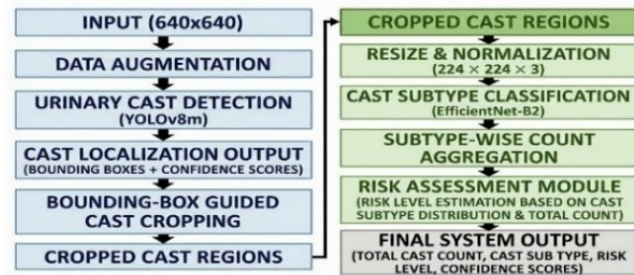


Fig. 4. Overview of the automated system for urinary cast detection, subtype classification and risk level estimation from microscopic images.

IV. RESULTS AND DISCUSSION

The proposed AI-based urine microscopy framework was evaluated using quantitative metrics provided in Tables I–VI for a variety of diagnostic tasks. Having everything considered, the findings show that carefully chosen lightweight deep learning models demonstrate excellent accuracy while preserving computational efficiency, making them appropriate for real-time clinical application.

Microscopy-based UTI analysis showed strong performance in detection, segmentation, and classification tasks, as highlighted in Table I. UNet++ with an EfficientNet-B0 encoder provided reliable spatial segmentation for WBCs and for yeast segmentation SegFormer-B2 (MiT-B2) model performed well. Meanwhile, YOLOv11-n enabled accurate and real-time WBC detection. Even when annotated bacterial bounding boxes were absent, EfficientNet-B0's classification-based method effectively distinguished particles. In assessing metadata-based models as additional decision-support tools, structured clinical indicators demonstrated significant discriminative power when used in a hierarchical fusion technique with microscopy evidence. However, prediction based solely on symptoms lacked reliability. Even though the symptom-based ML module demonstrated limited predictive power (ROC-AUC ≈ 0.57), it was included to explore multimodal fusion between clinical metadata and microscopy imaging. While the imaging pipeline provides the primary diagnostic signal, symptom data may enhance contextual interpretation in future multimodal models using late-fusion strategies.

TABLE I. PERFORMANCE OF MICROSCOPY-BASED MODELS FOR UTI ANALYSIS

Task	Model	Key Metrics
WBC Detection	YOLOv11-n	mAP@0.5 = 0.96
WBC Segmentation	UNet++ + EfficientNet-B0	Dice = 0.81, IoU ≈ 0.71
Yeast Detection	YOLOv11	mAP@0.5 > 0.94
Yeast Segmentation	SegFormer-B2 (MiT-B2)	Dice = 0.86, IoU = 0.75
Bacterial Classification	EfficientNet-B0	Validation Accuracy = 100%
Symptom-based ML	Classical ML	ROC-AUC ≈ 0.57
Clinical-Metadata ML	Classical ML	Near-perfect accuracy

YOLOv11-n was chosen for WBC detection because it had the highest overall mAP and F1-score while providing a good balance between precision and recall compared to Faster R-CNN. It is much smaller model size of 2.6 million parameters and very fast inference speed also make it better for efficient real-time use. UNet++ with EfficientNet-B0 outperformed the EfficientNet-B4 variation in WBC segmentation (Table II), achieving better Dice at 0.8053 and IoU at 0.7063. It also required fewer parameters and less memory, which indicates better generalization and suitability for deployment.

TABLE II. PERFORMANCE COMPARISON OF WBC DETECTION AND SEGMENTATION MODELS

Detection					
Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall	F1
YOLOv11-n	0.957	0.6139	0.8928	0.899	0.895
Faster R-CNN	0.952	0.5514	0.6730	0.978	0.796
Segmentation					
Model	Dice	IoU	Precision	Recall	F1
UNet++ (EfficientNet-B4)	0.7735	0.6697	0.8430	0.7936	0.7735

UNet++ (EfficientNet-B0)	0.805 3	0.7063	0.7997	0.8358	0.8107
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In yeast detection, RT-DETR-L had the highest absolute detection accuracy with a mean Average Precision (mAP@50) of 0.9581. However, YOLOv11 showed a better balance between accuracy and efficiency, making it a better choice for real-time clinical use (Table III). For yeast segmentation, SegFormer-B2 had the best Dice at 0.8565 and IoU at 0.7490 while using significantly fewer parameters than SegFormer-B4. This provides an optimal balance between accuracy and cost. This suggests it has better generalization and is more suitable for deployment. Overall, we considered the models' performance, stability, and ease of deployment within the proposed automated urine microscopy pipeline.

TABLE III. PERFORMANCE COMPARISON OF YEAST DETECTION AND SEGMENTATION MODELS

Detection					
Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall	F1
YOLOv11 (CSP-based)	0.9423	0.6120	0.9052	0.9016	0.9034
RT-DETR-L (ResNet Transformer)	0.9581	0.6457	0.9214	0.9189	0.9201
Segmentation					
Model	Dice	IoU	Precision	Recall	F1
SegFormer-B2(MiT-B2)	0.8565	0.7490	0.8481	0.8650	0.8565
SegFormer-B4(MiT-B4)	0.8524	0.7427	0.8309	0.8749	0.8524

Analyzing the results obtained through the Urine Crystals analysis (Table IV), it is important to learn the trade-offs between detection efficiency and accuracy. YOLO11m provided better properties concerning detection accuracy, speed, and computational cost, even as YOLOv8m had the best detection accuracy. EfficientNet-B0 had almost perfect results in classifying the crystals, except where there was a similarity in the shapes of the crystals.

TABLE IV. COMPARISON OF DEEP LEARNING MODELS FOR URINARY CRYSTAL DETECTION AND CLASSIFICATION IN URINE MICROSCOPY IMAGES

Detection				
Model	mAP@0.5	mAP@0.5:0.95	Precision	Recall
RT-DETR-L	0.888	0.556	0.854	0.885
YOLOv8m	0.915	0.599	0.849	0.865
YOLO11m	0.914	0.592	0.899	0.862
Classification				
Model	Test Accuracy		Macro F1	
EfficientNet-B0	99.41%		0.99	

RBC Morphology analysis (Table V) revealed consistent improvements in performance with an increase in model depth while performing RBC morphology analysis, reiterating the effectiveness of deeper EfficientNet variants. Additionally, object detection using YOLO, which helped isolate individual RBCs, improved overall classification accuracy, though ambiguity still plagues these datasets. Morphology-based analysis is employed to differentiate

dysmorphic RBCs. The improved EfficientNet-B3 model achieved a recall of 81%, indicating enhanced sensitivity in detecting clinically relevant RBC abnormalities compared to earlier results. This improvement strengthens the model's applicability for screening glomerular diseases. However, further optimization of recall remains important in clinical contexts, and future work will focus on techniques such as class-weighted loss functions and threshold tuning to further enhance sensitivity.

TABLE V. RBC MORPHOLOGY CLASSIFICATION PERFORMANCE

Model	Test Accuracy	Macro Precision	Macro Recall
EfficientNet-B0	62.89%	0.65	0.63
EfficientNet-B2	73.20%	0.75	0.73
EfficientNet-B3	81.40%	0.82	0.81

For urinary cast analysis (Table VI), the two-stage pipeline that combines YOLOv8m detection and EfficientNet-B2 classification showed strong performance even in difficult imaging conditions. While we effectively identified common cast types, the performance was lower for rarer subtypes due to class imbalance and similar shapes. Overall, these results confirm the strength and clinical importance of the suggested multi-task framework.

TABLE VI. URINARY CAST DETECTION AND CLASSIFICATION PERFORMANCE

Detection			
Model	mAP@0.5	mAP@0.5:0.95	Recall
YOLOv8m	0.801	0.495	0.738
Classification			
Model	Validation Accuracy	Macro F1	
EfficientNet-B2	74.48%	0.6254	

The integrated performance of the multi-module framework highlights the importance of task-specific model selection and fusion strategies, in addition to the good quantitative performance observed across the individual tasks. The results demonstrate that lightweight architectures may reach competitive or even state-of-the-art performance whilst maintaining computational efficiency and real-time deployment capabilities with appropriate optimization. The diagnostic modules, including UTI classification, kidney stone risk assessment, hematuria cause prediction, and cast-based renal disease analysis, may be employed independently whilst still providing a shared interpretation layer.

On the other hand, when the framework is compared with end-to-end black-box systems, there is an improvement with regards to transparency, whereby clinicians can relate the results with specific microscopic features, such as high WBC counts, dysmorphic RBCs, or crystals. Nevertheless, there were also several limitations, including low recall rates for dysmorphic RBC classification and poor performance with cast subtypes. For the improvement of the framework's performance, as well as its generalizability, there is a need to develop larger and more diverse datasets, as highlighted by the results. Despite the challenges faced, the framework shows high practicality, which can be used as a basis for future enhancements.

V. CONCLUSION

This study presents an AI-based framework for automated urine microscopy. It aims to reduce the variability, labor intensity, and scalability issues of manual urinalysis. The proposed system combines particle-level detection, segmentation, and classification with clinically informed decision logic to predict urinary tract and kidney diseases. Experimental results show strong performance across several tasks, including WBC, RBC, bacteria, yeast, crystal, and cast analysis, while keeping computational efficiency suitable for real-time clinical use. Unlike current methods that tackle these tasks separately, the proposed modular pipeline integrates microscopic evidence with clinical data. This improves diagnostic reliability, interpretability, and scalability. Future work will focus on expanding annotated datasets, improving bacteria localization, and validating the framework in real-world clinical settings.

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